

Preface

Who is This Book For?

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Why Write Yet Another Book About Linear Algebra?

The direct answer to this question is simple: I really think most resources are doing a *bad job* of explaining fundamental linear algebra to new students ¹. Not because they lack material or the complete picture - but because the common approach to teaching fundamental mathematical topics just doesn't work well for most people ².

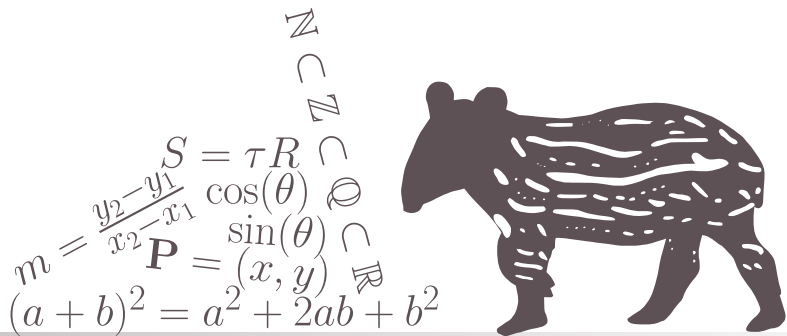
Introduction courses to fundamental mathematical topics usually look like this: here're some basic definitions, let's derive some specific results from them - now go apply these in your field.

¹and honestly, this isn't restricted to linear algebra but applies to all fundamental mathematical topics - but it's a classic case

²the main exception is students of pure mathematics and people who can easily deal with extremely abstract concepts - a very specific group of people at the end of the day

1

MATHEMATICAL BACKGROUND



It should be!

1.1 PREFACE

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Testing some equations: let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Then from the fact that $\mathbb{R}^n$ is a vector space, we get

$$\mathbf{c} = \alpha \mathbf{x} + \beta \mathbf{y} \in \mathbb{R}^n. \quad (1.1)$$

However, if $\mathbf{a} \in \mathbb{R}^m$, where $m \neq n$, then $\alpha \mathbf{x} \in \mathbb{R}^m$, and therefore $\mathbf{c}$ is not defined (i.e. Equation 1.1 is irrelevant).

Now, this is a test of the *aligned* environment, which allows for multiple (correctly-aligned) lines of equations and having the label only once (instead of per line, as in the *align* environment):

$$\begin{aligned} \sin(2x) &= \frac{1}{2i} \left(e^{2xi} - e^{-2xi} \right) \\ &= \frac{1}{2i} \left(\left[e^{xi} \right]^2 - \left[e^{-xi} \right]^2 \right) \\ &= \frac{1}{2i} \underbrace{\left(e^{xi} - e^{-xi} \right)}_{\sin(x)} \underbrace{\left(e^{xi} + e^{-xi} \right)}_{2 \cos(x)} \\ &= 2 \sin(x) \cos(x). \end{aligned} \quad (1.2)$$

Note 1.1 Test note

This is a test note. I will try to also refernce it later.



Now, I wish to reference [Note 1.1](#). Did it work?

Also, let us look at an example.

Example 1.1 Some test example

If $x = [1, 2, -3]$ and $y = [5, -1]$, then $n = 3$ and $m = 2$. Clearly, $n \neq m$ - and so $\alpha x \in \mathbb{R}^3$, while $\beta y \in \mathbb{R}^2$. This means that $\alpha x + \beta y$ is not defined.



...let's check if referencing an example works correctly: can you see [Example 1.1](#)?

I believe it is now time for some figures! See [Figure 1.1](#).

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

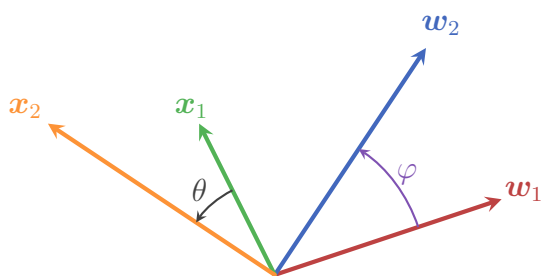


Figure 1.1: This is the first test figure.

Does it look ok?

sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus

CHAPTER 1. MATHEMATICAL BACKGROUND

dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

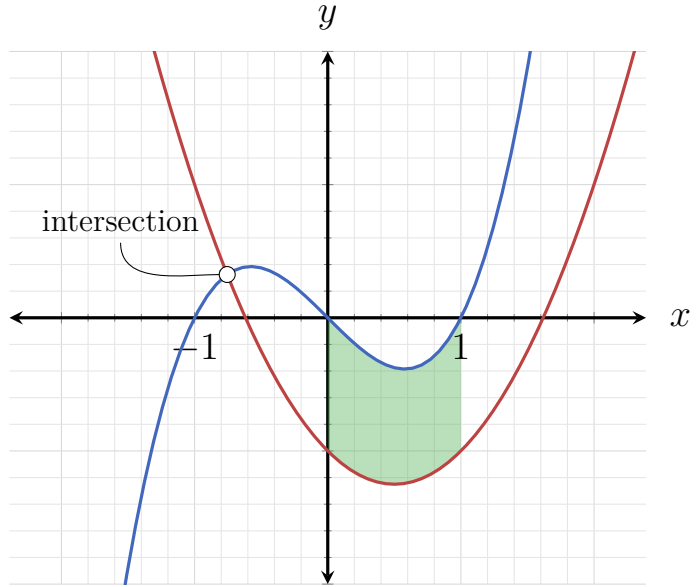


Figure 1.2: And this is another test figure! I hope it works.

Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellen-

1.1. PREFACE

tesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent

CHAPTER 1. MATHEMATICAL BACKGROUND

sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.1

$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$(\vec{A} \cdot \vec{B})^T = \vec{B}^T \cdot \vec{A}^T$$

$$\vec{A} \vec{x} = \lambda \vec{x}$$

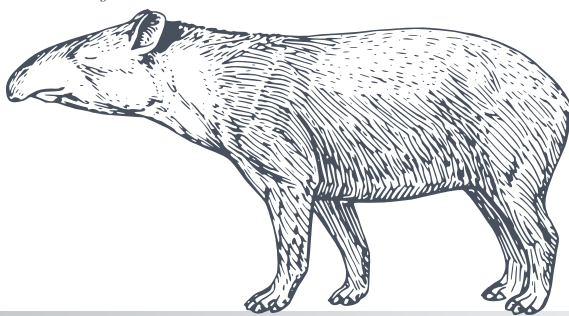
$$\det(\vec{A} \cdot \vec{B}) = \det(\vec{A}) \cdot \det(\vec{B})$$

$$\vec{C} = \vec{A} \cdot \vec{B} \Rightarrow c_j^i = \langle a^i | b_j \rangle$$

CHAPTER

2

THE BASICS OF LINEAR ALGEBRA



This chapter introduces the most important concepts of linear algebra, such as vector spaces, linear transformations, and eigenvectors and eigenvalues. Starting with simple 2- and 3-dimensional geometric concepts, the chapter guides the reader through the topics by introducing them in a visual and intuitive way, only generalizing them afterwards.

CHAPTER 2. THE BASICS OF LINEAR ALGEBRA

After reading this chapter and successfully completing the assignments at the end, the reader should have an intuitive understanding of basic linear algebra.

2.1 **VECTORS AND VECTOR SPACES**

2.2 LINEAR TRANSFORMATIONS

(HERE COMES LOTS OF INTRO TEXT ON LINEAR TRANSFORMATIONS WHICH I WILL WRITE LATER)

Linear transformations have a very nice property which makes applying them on different vectors extremely efficient. Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ which is a composition of scaling by a factor of 2 in the y -direction followed by a rotation of 90° anti-clockwise around the origin. Now let's say that we wish to calculate how T transforms the vector

$$\mathbf{a} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

In principle, we could simply apply the operations one after the other on $\mathbf{a}$:

$$\mathbf{a} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \xrightarrow{\times 2 \text{ in } y} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \xrightarrow{90^\circ \text{ rotation}} \begin{bmatrix} -4 \\ 3 \end{bmatrix} = T(\mathbf{a}). \quad (2.1)$$

Now, what happens if we wish to calculate a more complicated linear transformation, which is both composed out of many more basic transformations - and also acts on higher-dimensional vectors? Performing each transformation step might not be so straight-forward, and is definitely not an efficient process. In fact, even if we're dealing with a simple example just like the one we just saw, we would still need to apply it every time we want to calculate how a new vector is transformed.

It turns out that thanks to the simplicity of linear transformations¹ we only need to perform a few simple calculations per transformation to practically get any possible application of it on any vector!

To see how this is the case, let's go back to our simple 2-dimensional example. First, recall that we can express any vector in terms of the standard basis vectors (REF), which in this case gives us (note the color-coding)

$$\mathbf{a} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \overset{\substack{e_1 \\ \downarrow}}{3} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \overset{\substack{e_2 \\ \downarrow}}{2} \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (2.2)$$

¹more correctly: to the strict rules they enforce

2.2. LINEAR TRANSFORMATIONS

We can then apply the defining properties of linear transformations (REF) on $\mathbf{a}$:

$$\begin{aligned}
 T \begin{bmatrix} \textcolor{red}{3} \\ \textcolor{blue}{2} \end{bmatrix} &= T \left(\textcolor{red}{3} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \textcolor{blue}{2} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \\
 &= T \left(\textcolor{red}{3} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) + T \left(\textcolor{blue}{2} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \\
 &= \textcolor{red}{3} T \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \textcolor{blue}{2} T \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
 &= \textcolor{red}{3} T(\mathbf{e}_1) + \textcolor{blue}{2} T(\mathbf{e}_2).
 \end{aligned} \tag{2.3}$$

Now, applying T to the two standard basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ we get

$$\begin{aligned}
 \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} &\xrightarrow{\times 2 \text{ in } y} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \xrightarrow{90^\circ \text{ rotation}} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = T(\mathbf{e}_1) \\
 \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} &\xrightarrow{\quad} \begin{bmatrix} 0 \\ 2 \end{bmatrix} \xrightarrow{\quad} \begin{bmatrix} -2 \\ 0 \end{bmatrix} = T(\mathbf{e}_2)
 \end{aligned} \tag{2.4}$$

And when we use these transformed basis vectors in the linear combination we finally get

$$\begin{aligned}
 \textcolor{red}{3} T(\mathbf{e}_1) + \textcolor{blue}{2} T(\mathbf{e}_2) &= \textcolor{red}{3} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \textcolor{blue}{2} \begin{bmatrix} -2 \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} 0 \\ 3 \end{bmatrix} + \begin{bmatrix} -4 \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} -4 \\ 3 \end{bmatrix},
 \end{aligned} \tag{2.5}$$

which is exactly what we got when we calculated T 's effect directly on $\mathbf{a}$ (cf. [Equation 2.7](#)).

Notice exactly what happened: in the standard basis representation, $\mathbf{a}$ had the coefficients $\textcolor{red}{3}$ and $\textcolor{blue}{2}$ for $\mathbf{e}_1$ and $\mathbf{e}_2$, respectively. When applying T to $\mathbf{a}$ these two coefficients remained, but the basis changed: instead of $\{\mathbf{e}_1, \mathbf{e}_2\}$, the two basis vectors became $\{T(\mathbf{e}_1), T(\mathbf{e}_2)\}$.

CHAPTER 2. THE BASICS OF LINEAR ALGEBRA

Of course, this is not unique to $\mathbf{a}$. For example, let's look at another vector,

$$\mathbf{b} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}.$$

Its coefficients in the standard basis representation are 2 and -4, respectively. So we expect $T(\mathbf{b})$ to be

$$\begin{aligned} T(\mathbf{b}) &= 2T(\mathbf{e}_1) - 4T(\mathbf{e}_2) \\ &= 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 4 \begin{bmatrix} -2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 8 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 8 \\ 2 \end{bmatrix}. \end{aligned} \tag{2.6}$$

If we now calculate $T(\mathbf{b})$ directly, we get the following:

$$\mathbf{b} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} \xrightarrow{\times 2 \text{ in } y} \begin{bmatrix} 2 \\ -8 \end{bmatrix} \xrightarrow{90^\circ \text{ rotation}} \begin{bmatrix} 8 \\ 2 \end{bmatrix} = T(\mathbf{b}). \tag{2.7}$$

What this insight shows us is that given a linear transformation, we only need to know how it transforms the respective standard basis vectors in order to calculate how it transform any vector in its domain. The transformed version of the vector would simply be the linear combination of these transformed basis vectors, with the coefficients given by the vector's presentation in the standard basis. It really can't get any simpler than that!

To make this a bit more formal, let's look at the most generic case: say $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear transformation². Then applying T to a vector $\mathbf{v} \in \mathbb{R}^n$ in its standard basis representation,

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

²we'll deal with the more general case of $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ where $n \neq m$ in the next section.

2.2. LINEAR TRANSFORMATIONS

we get

$$\begin{aligned}
 T(\mathbf{v}) &= T \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \\
 &= T(v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_n \mathbf{e}_n) \\
 &= v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2) + \cdots + v_n T(\mathbf{e}_n).
 \end{aligned} \tag{2.8}$$

This is an incredibly powerful insight which allows us to save a lot of calculations - and which we will use in the next section to make the calculations of linear transformations *even more* concise and simple ³.

³spoiler: matrices

2.3 MATRICES

Following our example from the end of the previous section, let's look again at the result of applying T from Equation 2.7 on the vector $\mathbf{a} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$.

Since we know how T transforms the basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ (Equation 2.4), we can $T(\mathbf{a})$ in the following way:

$$T(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \overset{a_1}{\downarrow} \underset{\uparrow T(\mathbf{e}_1)}{\underset{3}{\color{red}}} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \overset{a_2}{\downarrow} \underset{\uparrow T(\mathbf{e}_2)}{\underset{2}{\color{blue}}} \begin{bmatrix} -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot \color{red}{3} - 2 \cdot \color{blue}{2} \\ 1 \cdot \color{red}{3} + 0 \cdot \color{blue}{2} \end{bmatrix}. \quad (2.9)$$

Then, we collect all the coefficients in black into a single structure, which we call a **matrix**:

$$M = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}. \quad (2.10)$$

Finally, we define the action of the matrix M on the vector $\mathbf{a}$ as following:

$$M\mathbf{a} = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \color{red}{3} \\ \color{blue}{2} \end{bmatrix} = \begin{bmatrix} 0 \cdot \color{red}{3} - 2 \cdot \color{blue}{2} \\ 1 \cdot \color{red}{3} + 0 \cdot \color{blue}{2} \end{bmatrix} = T(\mathbf{a}). \quad (2.11)$$

Of course, we should make sure that the matrix M we just defined works for other vectors as well. If we apply M to the second vector from our example, $\mathbf{b} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$, we get

$$M\mathbf{b} = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \color{red}{2} \\ \color{blue}{-4} \end{bmatrix} = \begin{bmatrix} 0 \cdot \color{red}{2} - 2 \cdot \color{blue}{-4} \\ 1 \cdot \color{red}{2} + 0 \cdot \color{blue}{-4} \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix} = T(\mathbf{b}), \quad (2.12)$$

which is exactly the result we got in Equation 2.6. Great!

Why not continue checking how this matrix-vector product we developed works for other vectors? First, let's check $\mathbf{e}_1$ and $\mathbf{e}_2$ for which we already know the results

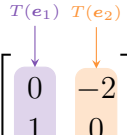
(Equation 2.7):

$$\begin{aligned} M\mathbf{e}_1 &= \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \textcolor{red}{1} \\ \textcolor{blue}{0} \end{bmatrix} = \begin{bmatrix} 0 \cdot \textcolor{red}{1} - 2 \cdot \textcolor{blue}{0} \\ 1 \cdot \textcolor{red}{1} + 0 \cdot \textcolor{blue}{0} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = T(\mathbf{e}_1), \\ M\mathbf{e}_2 &= \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \textcolor{red}{0} \\ \textcolor{blue}{1} \end{bmatrix} = \begin{bmatrix} 0 \cdot \textcolor{red}{0} - 2 \cdot \textcolor{blue}{1} \\ 1 \cdot \textcolor{red}{0} + 0 \cdot \textcolor{blue}{1} \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix} = T(\mathbf{e}_2), \end{aligned} \tag{2.13}$$

exactly the results we expected.

Perhaps you've noticed something by now - the columns of M are exactly the vectors $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$ in the right order:

$$M = \begin{bmatrix} \textcolor{purple}{0} & \textcolor{orange}{-2} \\ \textcolor{purple}{1} & \textcolor{orange}{0} \end{bmatrix}.$$

$\textcolor{purple}{T(\mathbf{e}_1)}$ $\textcolor{orange}{T(\mathbf{e}_2)}$


2.4 LINEAR EQUATIONS

2.5 **EIGENVECTORS AND EIGENVALUES**

2.6 MATRIX DECOMPOSITIONS

$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

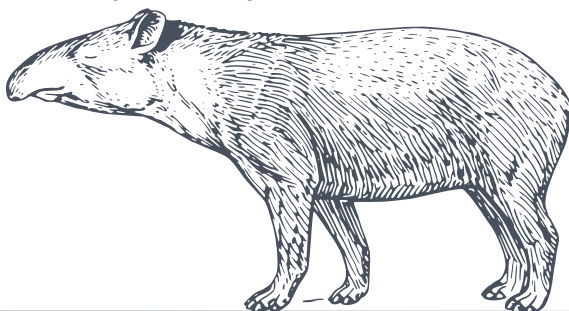
$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\mathbf{A} \vec{x} = \lambda \vec{x}$$

$$(\mathbf{A} \cdot \mathbf{B})^T = \mathbf{B}^T \cdot \mathbf{A}^T$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

3

INTRODUCING SOME FORMALISM

This is a test text for the Mathematical background chapter. Is this what's written in the actual chapter header? It should be!

$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

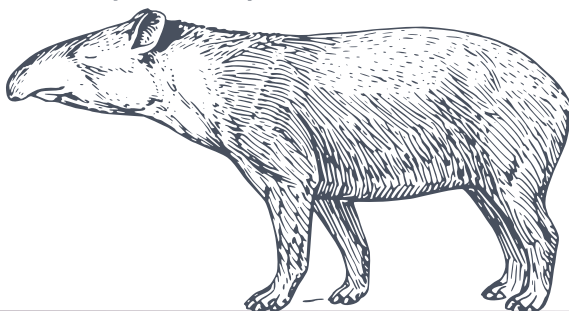
$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\mathbf{A} \vec{x} = \lambda \vec{x}$$

$$(\mathbf{A} \cdot \mathbf{B})^T = \mathbf{B}^T \cdot \mathbf{A}^T$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

4

ADVANCED TOPICS IN LINEAR ALGEBRA

This is a test text for the Mathematical background chapter. Is this what's written in the actual chapter header? It should be!

$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

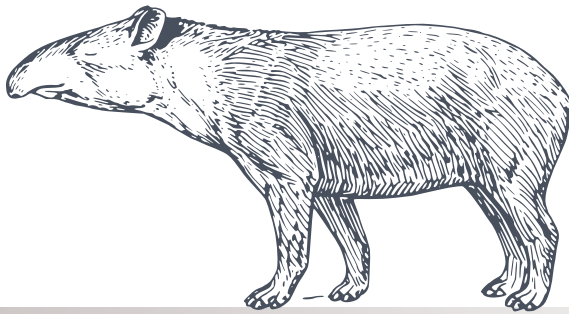
$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$(\vec{A} \cdot \vec{B})^T = \vec{B}^T \cdot \vec{A}^T$$

$$\mathbf{A} \vec{x} = \lambda \vec{x}$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

5

FURTHER RELATED TOPICS

This is a test text for the Mathematical background chapter. Is this what's written in the actual chapter header? It should be!

Actual page layout values.

CHAPTER 5. FURTHER RELATED TOPICS

<code>\paperheight = 845.04684pt</code>	<code>\paperwidth = 597.50787pt</code>
<code>\hoffset = 0.0pt</code>	<code>\voffset = 0.0pt</code>
<code>\evensidemargin = 35.28255pt</code>	<code>\oddsidemargin = -0.56833pt</code>
<code>\topmargin = -10.73834pt</code>	<code>\headheight = 18.125pt</code>
<code>\headsep = 21.75pt</code>	<code>\textheight = 591.5302pt</code>
<code>\textwidth = 418.25368pt</code>	<code>\footskip = 50.75pt</code>
<code>\marginparsep = 12.8401pt</code>	<code>\marginparpush = 6.52495pt</code>
<code>\columnsep = 10.0pt</code>	<code>\columnseprule = 0.0pt</code>
<code>1em = 12.0pt</code>	<code>1ex = 5.172pt</code>